

1. Naive approach to AUSM in RHD. The Method is exactly the same, only difference is Flux, H , and a are modified for the RHD case.

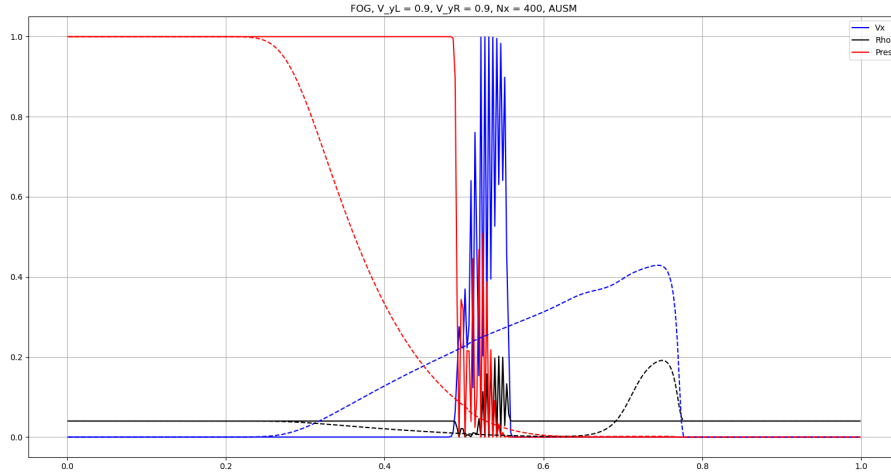


Figure 1: Dashed Lines are HLLC. So I don't think this is great

Ok, Fixed the problem. I forgot to remove the u_x from the flux. Resolving that the result looks better.

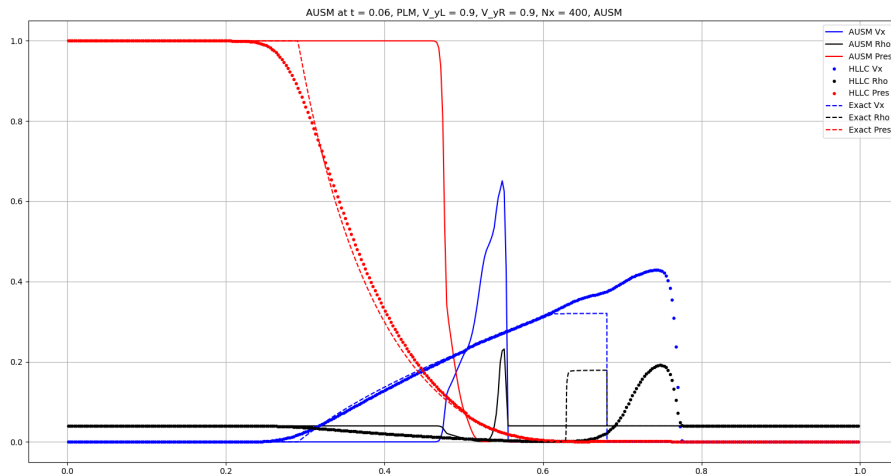


Figure 2: Good Start For AUSM at $T = 0.06$

Around $T = .14$ We get oscillations in AUSM which cause eventual failure. Solution is much better using FOG

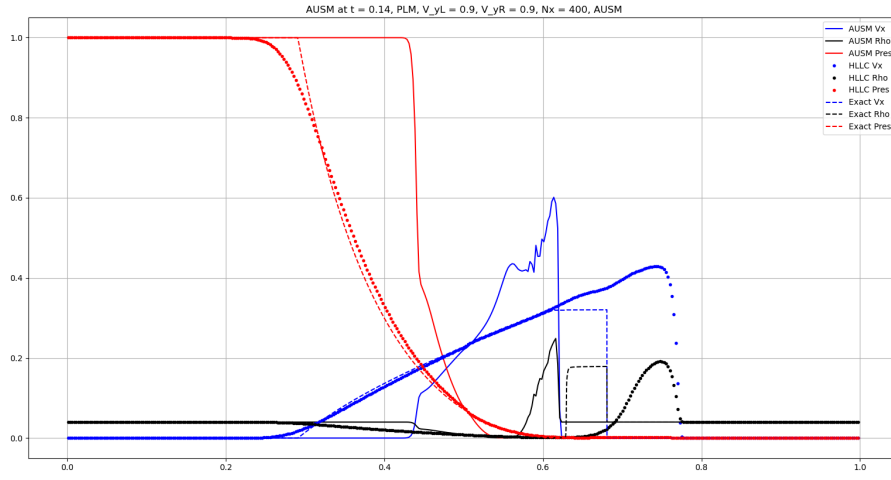


Figure 3: Wiggles show up at $T = 0.14$

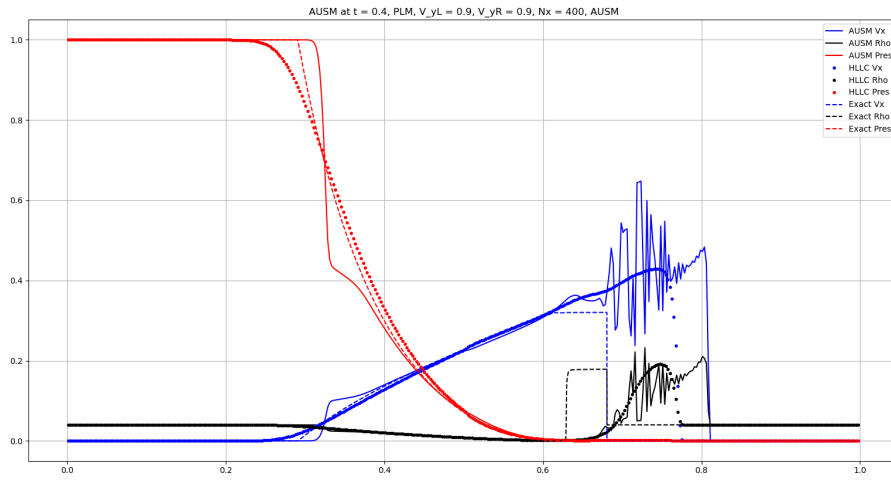


Figure 4: Ausm bad at $T = 0.4$

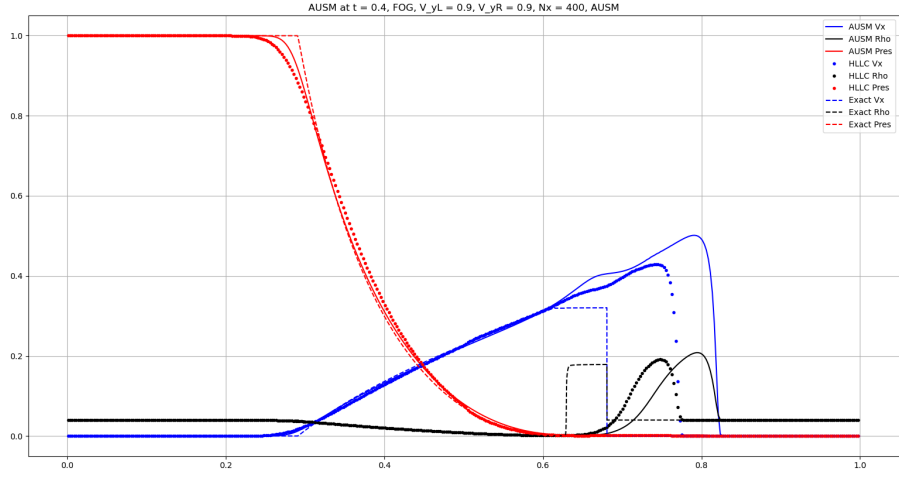


Figure 5: Disturbingly we see the shock moving even faster than with HLLC